



## 5255 - Defining the Dwarf Galaxy Reionization Transition Zone

Cycle: 3, Proposal Category: GO

### INVESTIGATORS

| <i>Name</i>                                    | <i>Institution</i>                           |
|------------------------------------------------|----------------------------------------------|
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### OBSERVATIONS

| <i>Folder</i> | <i>Observation</i> | <i>Label</i> | <i>Observing Template</i> | <i>Science Target</i> |
|---------------|--------------------|--------------|---------------------------|-----------------------|
| SagDIG        |                    |              |                           |                       |
|               | 1                  | SagDIG       | NIRCam Imaging            | (1) SagDIG            |

### ABSTRACT

The impact of reionization on galaxies is relatively well understood in a dichotomous sense: massive galaxies are not impacted by reionization and low-mass galaxies have their star formation terminally extinguished. Nonetheless, there is an intermediate regime where the star formation in relatively low-mass galaxies is impacted by heating during reionization but can recover and continue at a later date. We know of this "transition zone" from recovered star formation histories of three nearby isolated currently star-forming galaxies. With JWST we can add a fourth galaxy to this

sample (SagDIG) which has only shallow HST photometry primarily due to high foreground extinction. Post-reionization re-ignited star formation in dwarf galaxies has implications for cosmological simulations. Specifically, if delayed star formation is typical for lower galaxy masses, this would significantly impact the interpretation that low-mass galaxies are engines of reionization and, thus, our expectations of the high-redshift galaxy UV luminosity function. JWST observations of SagDIG will help us to better understand this important galaxy mass regime with regard to the effects of reionization.

## **OBSERVING DESCRIPTION**

We will image SagDIG with NIRCcam with the short-wavelength filters F090W and F150W, with simultaneous imaging in the long-wavelength filters F250M and F480M. Our science goals require that the SW imaging reach below the old main sequence turn-off with a SNR~10 for a metal-poor, 13 Gyr old star. We have used a MEDIUM8 readout and a subpixel dither pattern to mitigate cosmic rays and bad pixels, improve PSF sampling, and reduce flat-field errors thereby maximizing the SNR. We do not intend to cover the chip gaps as uniform depth is preferred for the imaged stellar field. We have selected 8 groups per integration, 2 integrations, and 12 dithers per filter. A total science time of ~11 hrs is needed to achieve our photometric depth requirement. Total requested time with overhead from the APT is ~14 hours. We impose a background limited constraint of 10% as the background is highly variable in this region of the sky. We selected additional orient constraints to ensure the NIRCcam field of view optimally covers the stellar disk of the galaxy while avoiding any bright nearby stars.

# Proposal 5255 - Targets - Defining the Dwarf Galaxy Reionization Transition Zone

| Fixed Targets | #                                                                                                                                                                                   | Name   | Target Coordinates                                                                  | Targ. Coord. Corrections | Miscellaneous |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------|--------------------------|---------------|
|               | (1)                                                                                                                                                                                 | SagDIG | RA: 19 30 0.4871 (292.5020296d)<br>Dec: -17 42 17.45 (-17.70485d)<br>Equinox: J2000 | Epoch of Position: 2000  |               |
|               | <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=Galaxy</i><br><i>Description=[Dwarf irregular galaxies]</i> |        |                                                                                     |                          |               |

# Proposal 5255 - Observation 1 - Defining the Dwarf Galaxy Reionization Transition Zone

|                      |                                                                                                                                                                                                                                                                                           |                     |                                                                                     |                 |            |                          |                    |               |                     |                    |                              |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|-----------------|------------|--------------------------|--------------------|---------------|---------------------|--------------------|------------------------------|
| Observation          | Proposal 5255, Observation 1: SagDIG                                                                                                                                                                                                                                                      |                     |                                                                                     |                 |            |                          |                    |               |                     |                    | Fri Mar 01 01:05:24 GMT 2024 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                                                                                                |                     |                                                                                     |                 |            |                          |                    |               |                     |                    |                              |
|                      | Observing Template: NIRCcam Imaging                                                                                                                                                                                                                                                       |                     |                                                                                     |                 |            |                          |                    |               |                     |                    |                              |
|                      | Comments: We request that these observations not be taken directly following observations of a very bright target. From the ERS program, persistance can impact the precise photometric recovery of faint stars, which is the primary goal of this observing program (Weisz et al. 2023). |                     |                                                                                     |                 |            |                          |                    |               |                     |                    |                              |
| Diagnostics          | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                               |                     |                                                                                     |                 |            |                          |                    |               |                     |                    |                              |
| Fixed Targets        | #                                                                                                                                                                                                                                                                                         | Name                | Target Coordinates                                                                  |                 |            | Targ. Coord. Corrections |                    |               | Miscellaneous       |                    |                              |
|                      | (1)                                                                                                                                                                                                                                                                                       | SagDIG              | RA: 19 30 0.4871 (292.5020296d)<br>Dec: -17 42 17.45 (-17.70485d)<br>Equinox: J2000 |                 |            | Epoch of Position: 2000  |                    |               |                     |                    |                              |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                                                                                                                                         |                     |                                                                                     |                 |            |                          |                    |               |                     |                    |                              |
|                      | Category=Galaxy<br>Description=[Dwarf irregular galaxies]                                                                                                                                                                                                                                 |                     |                                                                                     |                 |            |                          |                    |               |                     |                    |                              |
| Template             | Module                                                                                                                                                                                                                                                                                    |                     | Subarray                                                                            |                 |            |                          | Target Placement   |               |                     |                    |                              |
|                      | ALL                                                                                                                                                                                                                                                                                       |                     | FULL                                                                                |                 |            |                          | Module Gap         |               |                     |                    |                              |
| Dithers              | #                                                                                                                                                                                                                                                                                         | Primary Dither Type |                                                                                     | Primary Dithers |            | Subpixel Dither Type     |                    | Dither Size   |                     | Subpixel Positions |                              |
|                      | 1                                                                                                                                                                                                                                                                                         | NONE                |                                                                                     |                 |            | STANDARD                 |                    |               |                     | 12                 |                              |
| Spectral Elements    | #                                                                                                                                                                                                                                                                                         | Short Filter        | Long Filter                                                                         | Readout Pattern | Groups/Int | Integrations/Exp         | Total Integrations | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID   |                              |
|                      | 1                                                                                                                                                                                                                                                                                         | F090W               | F250M                                                                               | MEDIUM8         | 8          | 2                        | 24                 | 12            | 20228.075           |                    |                              |
|                      | 2                                                                                                                                                                                                                                                                                         | F150W               | F480M                                                                               | MEDIUM8         | 8          | 2                        | 24                 | 12            | 20228.075           |                    |                              |
| Special Requirements | Aperture PA Range 77.92542306 to 88.92542306 Degrees (V3 78.0 to 89.0)<br>Background Limited. Background no more than 10th percentile above minimum                                                                                                                                       |                     |                                                                                     |                 |            |                          |                    |               |                     |                    |                              |